The Recovery–Certification Gap: A Theory of Adversarial Unlearning

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Abstract

Machine unlearning is supposed to remove specific training data, and methods are increasingly shipped with *certificates* that the data is gone. This paper makes one claim: unlearning robustness reduces to a single number an auditor can measure with no retrained reference model—the per-example *recovery radius*, the smallest input change that makes the model produce a forgotten class again—and that number, not clean forget accuracy, is what a certificate must answer to. I develop this through four theorems, each matched to experiments on an eight-method CIFAR-10 benchmark (SalUn, SCRUB, RURK, CE-U, SGA, BalDRO, robust-base SalUn, and ORBIT—a repo-defined internal stress-test row, not an external baseline). (1) A *recovery–certification gap*: a method's measured recovery radius implies a floor that any certified-unlearning budget must respect, so an oracle-free audit can already challenge an over-tight certificate. (2) A *selective-leakage test* calibrated against a zero-knowledge oracle null — a model retrained without the forgotten class. The null carries no intrinsic forget-vs-retain asymmetry, so a gap over it is genuine residual knowledge: SalUn and the smoothed objective leak, while SCRUB's apparent leakage turns out to be seed-fragile. (3) A *covering-family impossibility*: worst-case robustness to all patch and border-frame delivery surfaces at once forces a constant classifier, which is why mixed-surface patch defenses cannot transfer. (4) A Gaussian-smoothed *certified-recovery objective*, to my knowledge the first to attach a randomized-smoothing certificate to unlearning; across three seeds it drives forget accuracy to 9–12% with retain accuracy preserved and certifies an L_2 radius of 0.193 that, measured in its own norm, sits safely below the radius at which the class re-emerges (0.46–0.57). Sweeping the smoothing scale exposes a hard certification–utility tradeoff: a larger certificate is bought with weaker forgetting. The multi-regime benchmark becomes the evidence for a single thesis—audit unlearning by recovery radius, not by clean accuracy. Code and checkpoints are available at `https://github.com/DevT02/ForgetGate`.

1 Introduction

Machine unlearning is supposed to remove specific training data from a trained model, and a growing line of work issues *certificates* that the data is gone. Most evaluations still check removal with clean forget accuracy—does the model still predict the forgotten class on unperturbed inputs? This paper takes a different position: clean accuracy is the wrong test, and the right one is a single measurable quantity. If a small, targeted perturbation re-elicits a forgotten class, the information was never really erased; so I measure the *recovery radius*—the smallest such perturbation—and argue it is the

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quantity that both hands-on red-team audits and formal unlearning certificates are really about. A method can pass clean-accuracy and even rate-based audits and still leak under it.

I evaluated eight unlearning methods under four attack families—per-example conditional recovery, universal perturbations, conditional patches, and border-frame and multi-patch variants—on a controlled CIFAR-10 benchmark. The short version is that no single attack breaks everything, but no single method is safe either. Different methods fail in different ways. Some are vulnerable to targeted per-example recovery but not patches. Some look robust until you try a slightly different attack surface. And a defense that closes one hole often leaves another open.

That framing raises a harder question than simple attack success: when a perturbation recovers the forgotten class, is it actually exposing residual class-specific information that the unlearning method failed to erase, or is it merely inducing a generic adversarial target-label failure? The paper is designed to separate those cases. I therefore compare forget-set examples against matched retain-class controls under the same targeted recovery audit, and—to rule out the chance that forget inputs are merely intrinsically closer to the target class—against a zero-knowledge oracle null, a model retrained without the forgotten class. If forget examples are selectively easier to push back to the forgotten class than this null, that is genuine residual knowledge. If forget and retain controls are equally easy to hijack, the result is better understood as generic adversarial fragility than as recovery of supposedly forgotten information.

The positive results are modest but real: a post-hoc feature-subspace defense modestly increases recovery radii on BalDRO, SalUn, and ORBIT without hurting clean accuracy (RURK and SCRUB are essentially unaffected), and patch-aware training partially closes the patch channel on ORBIT and CE-U—though it does not transfer to frame or multi-patch delivery. The benchmark, code, and checkpoints are fully reproducible at `https://github.com/DevT02/ForgetGate`.

These observations are not a grab-bag: all four attack regimes measure one quantity—the per-example recovery radius—and that quantity is what ties red-team auditing to certified unlearning, developed as a small theory in Section 3. Four theorems organize the narrative. Theorem 1 (recovery–certification gap) bounds the recovery radius below by the target margin and a Lipschitz constant, so that any claimed certified-unlearning budget implies a minimum recovery radius an oracle-free audit can check against—a necessary condition rather than a strict falsification, since the Lipschitz term is estimated by a local proxy. Theorem 2 (Pinsker selective leakage) recasts the forget-vs-retain control gap as a variational-information test, calibrated against a zero-knowledge oracle null, that distinguishes genuine residual-knowledge recovery from generic adversarial fragility. Theorem 3 (local-delivery transfer impossibility) proves that worst-case robustness to a covering family of delivery surfaces forces constancy, which converts the paper's "mixed-surface stage-2 is future work" open problem into a negative result. Theorem 4 (smoothed certified-recovery objective) replaces the inner attacker of the proposed defense with Gaussian smoothing and inherits a randomized-smoothing certificate, giving, to my knowledge, the first machine-unlearning objective with a per-input certified lower bound on recovery radius. Section 8 then validates each theorem against the benchmark: the Lipschitz–radius scatter (Thm 1), the Pinsker KL table (Thm 2), the patch/frame/multi-patch transfer failure (Thm 3), and an end-to-end predicted-vs-measured certified radius (Thm 4).

2 Threat Models and Attack Regimes

2.1 Universal perturbations

The first regime includes Fourier, tile, warp, and codebook-style universal perturbations. These attacks are useful primarily as a negative result: on stronger methods, the residual vulnerability is not well-compressed into one shared low-complexity direction.

2.2 Per-example conditional recovery

The main broad attack optimizes the forgotten-class target margin

$$m_t(x) = z_t(x) - \max_{j \neq t} z_j(x), \tag{1}$$

using strong first-order optimizers such as `adam_margin` and `apgd_margin`. This regime asks whether a forgotten behavior is locally recoverable by a targeted per-example perturbation inside a fixed norm budget.

2.3 Amortized short-budget attacks

I additionally study a learned image-conditional perturbation generator trained from short Adam teacher trajectories and refined with a tiny local budget. This attack family is not a universal win, but it improves matched low-step attacks on ORBIT and SalUn.

2.4 Conditional patches

I also study a conditional small-area patch regime in which both patch content and placement depend on the input. This is a patch-based regime, not a universal trigger claim.

2.5 Additional local delivery surfaces

To test whether patch-aware defenses generalize beyond one square patch, I also study conditional border-frame attacks and conditional multi-patch attacks. These are stronger local-delivery checks than a single-patch evaluation because they alter geometry while preserving the same selective red-team objective.

3 A Theory of Adversarial Unlearning

The empirical regimes of this paper measure a single object: the smallest input-space perturbation that re-elicits the forgotten class on a per-example basis. This section names that object, ties it quantitatively to certified unlearning, separates selective leakage from generic adversarial fragility, and gives a constructive impossibility result for the open "local-delivery transfer" problem. The benchmark sections that follow are re-cast as empirical validation of these statements rather than as standalone attack curves.

3.1 Setup and the residual forget-channel capacity

Notation. Let $\mathcal{X} = [0,1]^d$ be the input space and $\mathcal{Y} = [K]$ the label set. Fix the forget class $t \in [K]$. Let $D = D_R \cup D_F$ with $D_F = \{(x_i, t)\}_{i=1}^{n_f}$. Let $\mathcal{A}$ be an unlearning algorithm producing $\theta_u = \mathcal{A}(D, t)$. Let θ_o denote a from-scratch oracle trained on D_R. Let $f_\theta : \mathcal{X} \to \mathbb{R}^K$ be the logit map and define the forget-class *target margin*

$$m_t(x;\theta) \;:=\; f_\theta(x)_t - \max_{j \neq t} f_\theta(x)_j. \tag{2}$$

The unlearned model predicts the forgotten class at x iff $m_t(x;\theta_u) > 0$.

Recovery radius. For an L_∞ budget,

$$r_t(x;\theta) \;:=\; \inf\bigl\{\epsilon \ge 0 \mid \exists\, x' \in \mathcal{X}, \|x'-x\|_\infty \le \epsilon,\; m_t(x';\theta) > 0\bigr\}, \tag{3}$$

with $r_t(x;\theta) = 0$ when the clean prediction already matches. A per-example audit returns a finite-budget estimator $\hat{r}_t(x;\theta_u)$ via probe-then-binary search with PGD / APGD inner loops.

The right population object. Let μ_t denote the forget-class input distribution and $\mu_{\neg t}$ the matched non-forget control distribution used throughout the paper. Define the *residual forget-channel capacity* at scale ϵ:

$$\boxed{\mathcal{R}_\epsilon(\theta_u) \;:=\; \Pr_{x\sim\mu_t}\bigl[r_t(x;\theta_u) \le \epsilon\bigr] \;-\; \Pr_{x\sim\mu_{\neg t}}\bigl[r_t(x;\theta_u) \le \epsilon\bigr].} \tag{4}$$

$\mathcal{R}_\epsilon \in [-1,1]$ is the signed difference between forget-side and retain-side recoverability at audit scale ϵ. The forget–retain gap reported throughout the paper is exactly the empirical estimator of $\mathcal{R}_\epsilon$. Two qualitative regimes the paper invokes informally become precise statements about $\mathcal{R}_\epsilon$:

- **Selective leakage** (paper's ORBIT/CE-U story): $\mathcal{R}_\epsilon(\theta_u) > 0$ — forget inputs are strictly easier to recover than retain controls.
- **Generic fragility** (paper's SCRUB story): $\mathcal{R}_\epsilon(\theta_u) \approx 0$ — the model is adversarially fragile, but not selectively so on the forget class.

3.2 Theorem 1: the recovery–certification gap

I bridge the per-example recovery radius to any *certified-unlearning* budget the method might claim.

Theorem 1 (Recovery–certification gap)**.** *Suppose the target-margin function is L-Lipschitz in $\|\cdot\|_\infty$ over $\mathcal{X}$, i.e. $|m_t(x;\theta_u) - m_t(x';\theta_u)| \le L\,\|x-x'\|_\infty$ for all $x, x' \in \mathcal{X}$. Then for every $x \in \mathcal{X}$,*

$$r_t(x;\theta_u) \;\ge\; \frac{-m_t(x;\theta_u)}{L}. \tag{5}$$

Equivalently,

$$\Pr_{x\sim\mu_t}\bigl[r_t(x;\theta_u) \le \epsilon\bigr] \;\le\; \Pr_{x\sim\mu_t}\bigl[\,m_t(x;\theta_u) \ge -L\epsilon\,\bigr]. \tag{6}$$

Proof. For any $\epsilon \ge 0$ and any $x' \in B_\infty(x,\epsilon)\cap\mathcal{X}$, the Lipschitz bound gives $m_t(x';\theta_u) \le m_t(x;\theta_u) + L\epsilon$. For the recovery event $m_t(x';\theta_u) > 0$ to hold we therefore require $\epsilon > -m_t(x;\theta_u)/L$, which proves (5). Then (6) follows by integrating against μ_t. □

Corollary 1 (Certified unlearning needs separated margins)**.** *Instantiate Theorem 1 with the* probability margin $\bar{m}_t(x;\theta) := \pi_\theta(x)_t - \max_{j\ne t}\pi_\theta(x)_j \in [-1,1]$, *where* $\pi_\theta = \mathrm{softmax}\circ f_\theta$. *Because softmax is strictly monotone, $\bar{m}_t$ has the same sign as the logit margin and defines the same recovery event; let L be its input-space Lipschitz constant. Let θ_u be (ϵ_c,δ)-certified relative to the oracle θ_o in the conditional TV sense,*

$$\Pr_{x\sim\mu_t}\Bigl[\mathrm{TV}\bigl(\pi_{\theta_u}(\cdot\mid x),\, \pi_{\theta_o}(\cdot\mid x)\bigr) \le \epsilon_c\Bigr] \;\ge\; 1-\delta,$$

and suppose the oracle is separated on average, $\mathbb{E}_{\mu_t}\bigl[-\bar{m}_t(x;\theta_o)\bigr] \ge \gamma > 0$. Then

$$\mathbb{E}_{x\sim\mu_t}\bigl[\,r_t(x;\theta_u)\,\bigr] \;\ge\; \frac{\gamma - 2(\epsilon_c+\delta)}{L}. \tag{7}$$

Proof. For distributions p, q on $[K]$ and any class k, the event $A = \{k\}$ gives $|p_k - q_k| = |p(A) - q(A)| \leq \mathrm{TV}(p, q)$, and $a \mapsto \max_{j \neq t} a_j$ is 1-Lipschitz in the sup-norm. Hence pointwise,

$$\begin{aligned} |\bar{m}_t(x; \theta_u) - \bar{m}_t(x; \theta_o)| &\leq |\pi_{\theta_u}(x)_t - \pi_{\theta_o}(x)_t| + \big|\max_{j \neq t} \pi_{\theta_u}(x)_j - \max_{j \neq t} \pi_{\theta_o}(x)_j\big| \\ &\leq 2\,\mathrm{TV}\big(\pi_{\theta_u}(\cdot \mid x), \pi_{\theta_o}(\cdot \mid x)\big). \end{aligned}$$

Since the TV distance is bounded by ϵ_c with probability $1 - \delta$ and by 1 otherwise, its expectation is at most $\epsilon_c(1 - \delta) + \delta \leq \epsilon_c + \delta$. Taking expectations under μ_t gives $\mathbb{E}[-\bar{m}_t(\cdot; \theta_u)] \geq \gamma - 2(\epsilon_c + \delta)$; applying Theorem 1 to $\bar{m}_t$ (pointwise $r_t(x) \geq -\bar{m}_t(x)/L$) yields the claim. The bounded margin $\bar{m}_t \in [-1, 1]$ removes any dependence on a logit bound. □

What this says about the benchmark. Every method in the paper has a measured $\hat{r}$ distribution and a straightforwardly estimable Lipschitz proxy $\hat{L}$ (e.g. the spectral norm of the Jacobian of the margin at audit points, which an existing Jacobian-access audit already computes), used here as a logit-margin proxy for the probability-margin constant L of Corollary 1. Corollary 1 contrapositively gives a *theoretical floor on the certifiable unlearning budget*:

$$\epsilon_c + \delta \;\geq\; \frac{\gamma - L \cdot \mathbb{E}[\hat{r}]}{2},$$

estimated per method from its recovery-radius audit; the scatter of Section 8 plots the more robust median $\hat{r}$ as a stand-in for the mean. Two gaps separate $\hat{L}$ from the constant Corollary 1 requires. First, a *norm* gap: the L_∞-Lipschitz constant of the scalar margin is controlled by the dual norm $\|\nabla m\|_1$, whereas the spectral Jacobian proxy reports $\|\nabla m\|_2 \leq \|\nabla m\|_1$ and can therefore understate it by up to a $\sqrt{d}$ factor; a tighter audit would estimate $\|\nabla m\|_1$ directly. Second, a *locality* gap: $\hat{L}$ is a local proxy for the (NP-hard) global Lipschitz constant, and local proxies on ViTs tend to under-estimate it. Both gaps push the floor the same way—they inflate it and could in principle produce a false rejection—so I treat this as an audit-estimable *necessary condition* rather than a strict falsification test: a method whose median recovery radius is small relative to the oracle margin is unlikely to be honestly marketable at small ϵ_c, and the bound serves as a heuristic ranking and baseline reality-check that audits with no access to the oracle distribution can already perform. This converts the paper's empirical numbers into a stake in the certified-unlearning literature [10, 6, 21, 19]: a modern data-removal certificate such as the retain-sensitivity calibration of [19] supplies exactly the (ϵ_c, δ) budget that Corollary 1 turns into a recovery-radius lower bound.

3.3 Theorem 2: selective leakage is a Pinsker test

Theorem 2 (Pinsker bound on selective leakage)**.** *Let* $p_F := \Pr_{\mu_t}[r_t \leq \epsilon]$ *and* $p_R := \Pr_{\mu_{\neg t}}[r_t \leq \epsilon]$*. Then*

$$|\mathcal{R}_\epsilon(\theta_u)| \;=\; |p_F - p_R| \;\leq\; \sqrt{\tfrac{1}{2}\,\mathrm{KL}\big(\mathrm{Ber}(p_F)\,\|\,\mathrm{Ber}(p_R)\big)}. \tag{8}$$

Consequently, $|\mathcal{R}_\epsilon(\theta_u)| \geq c$ *forces* $\mathrm{KL}\big(\mathrm{Ber}(p_F)\|\mathrm{Ber}(p_R)\big) \geq 2c^2$*, i.e. the audit's binary recoverability outcome distinguishes forget from retain at variational rate* c*.*

Proof. Pinsker's inequality applied to the two Bernoulli laws on the binary outcome $\mathbf{1}\{r_t \leq \epsilon\}$. □

Corollary 2 (SCRUB is a generic-fragility regime)**.** *If* $\mathcal{R}_\epsilon(\theta_u) \to 0$ *as* $n_f \to \infty$ *at some audit scale* ϵ*, no unlearning claim about* μ_t*-specific information removal is empirically supported at that scale: the data-conditional binary leakage event is indistinguishable from the retain-control baseline.*

Why this matters. The paper's qualitative observation that SCRUB "exhibits generic adversarial fragility rather than selective forgetting leakage" becomes a variational-information comparison (not a finite-sample hypothesis test; see the small-n caveat in the limitations): the binary recoverability outcomes on forget and retain controls are samples from $\mathrm{Ber}(p_F)$ and $\mathrm{Ber}(p_R)$, and Theorem 2 turns an observed gap into a *lower* bound on the KL it carries about the forget condition. The direction matters: a large measured gap (ORBIT/CE-U) *forces* a quantitative KL and is genuine selective signal, whereas a near-zero gap (SCRUB) leaves the KL unconstrained from above, so the generic-fragility reading rests on the direct equality of the measured rates, not on Pinsker.

3.4 Theorem 3: local-delivery transfer impossibility

This formalizes the paper's open problem ("patch-aware defenses fail to transfer to frame or multi-patch delivery") as a structural impossibility.

Delivery families and coordinate coverage. A *delivery mask* is a subset $M \subseteq [d]$ of input coordinates. A *delivery family* $\mathcal{D} \subseteq 2^{[d]}$ is *coordinate-covering* if every input coordinate lies in at least one of its masks, $\bigcup_{M \in \mathcal{D}} M = [d]$. A delivery-conditional perturbation under $\mathcal{D}$ with budget ϵ is a $\delta \in \mathbb{R}^d$ with $\mathrm{supp}(\delta) \subseteq M$ for some $M \in \mathcal{D}$ and $\|\delta\|_\infty \leq \epsilon$.

Lemma 1 (Patches alone are coordinate-covering)**.** *For 224×224 images on $[0,1]^d$, the family of all 32×32 square delivery masks at every position is coordinate-covering; adjoining the 16-pixel border frames leaves it coordinate-covering.*

Proof. Every pixel lies inside at least one 32×32 window, so the union of the patch masks is all of $[d]$; adding more masks only enlarges the union. □

Theorem 3 (Budget-graded local-delivery transfer impossibility)**.** *Let $\mathcal{D}$ be a coordinate-covering delivery family and let $f_\theta : [0,1]^d \to \mathcal{Y}$ be any classifier. Fix* any *budget $\epsilon > 0$ and suppose f_θ is* worst-case invariant *to all delivery-conditional perturbations under $\mathcal{D}$ at budget ϵ:*

$$\forall x,\ \forall M \in \mathcal{D},\ \forall \delta \text{ with } \mathrm{supp}(\delta) \subseteq M,\ \|\delta\|_\infty \leq \epsilon,\ x + \delta \in [0,1]^d : \quad f_\theta(x+\delta) = f_\theta(x). \tag{9}$$

Then f_θ is constant on $[0,1]^d$.

Proof. Fix $x, x' \in [0,1]^d$; we connect them by admissible ϵ-steps, one coordinate at a time. For coordinate i, choose a mask $M_i \in \mathcal{D}$ with $i \in M_i$ (coordinate coverage). Move the i-th entry monotonically from its current value to x'_i in increments of magnitude at most ϵ; each increment is a perturbation supported on $\{i\} \subseteq M_i$ with sup-norm $\leq \epsilon$, and every intermediate point stays in $[0,1]^d$. Worst-case invariance leaves f_θ unchanged at each increment, so resetting coordinate i from x_i to x'_i does not change f_θ. After d coordinates the point equals x', hence $f_\theta(x) = f_\theta(x')$. As x, x' were arbitrary, f_θ is constant. The number of steps, $\sum_i \lceil |x'_i - x_i|/\epsilon \rceil$, is finite for every $\epsilon > 0$. □

Corollary 3 (No-free-lunch for stage-2 patch defense)**.** *For* any *budget $\epsilon > 0$, no fine-tune of a non-constant classifier on $[0,1]^d$ can be worst-case invariant on a coordinate-covering delivery family (e.g. all 32×32 patches, with or without 16-pixel border frames) while keeping clean accuracy strictly above chance. In particular the impossibility holds at the small budgets $\epsilon \approx 8/255$ the audits use, not only in a large-budget limit.*

What this says about the open problem. Worst-case multi-surface robustness against a coordinate-covering delivery family is functionally equivalent to input-invariance, which forces the classifier to be constant *at every budget*—the chaining argument needs only that each coordinate be reachable, so the obstruction is present already at the $\epsilon \approx 8/255$ the audits use, not merely in a large-budget limit. The empirical failure of *mixed-surface stage-2* reported in the defense and validation sections of this paper is not an engineering artifact: it is the only thing that can happen for a non-trivial classifier under worst-case multi-surface training. The constructive consequence is that "local-delivery transfer" should be retargeted from worst-case invariance to *realistic delivery distributions* — i.e. stochastic mask priors that are *not* covering. This paper implements one such relaxation by sampling patch positions rather than taking a worst case over them; I name this a *ρ-thin delivery defense* and treat it as the appropriate target for local-delivery robustness, since it is the only version of the problem not foreclosed by Theorem 3.

3.5 Theorem 4: a smoothed certified-recovery objective

This paper's proposed objective (13) is a targeted-margin adversarial unlearning objective with inner-loop attack $\delta^*(x)$. I upgrade it to a *certified* counterpart by replacing the inner loop with Gaussian smoothing of the margin penalty, and prove the resulting objective enjoys a randomized-smoothing certificate inherited from the Cohen-Rosenfeld-Kolter framework on the smoothed margin function.

Smoothed objective. Let $\phi(z) = \max\{0, z\}$ and fix $\sigma > 0$.

$$\mathcal{L}_\sigma(\theta) \;:=\; \mathbb{E}_{x\sim D_R}\big[L_{\text{ret}}(x;\theta)\big] \;+\; \lambda_f\,\mathbb{E}_{x\sim D_F}\big[L_{\text{forg}}(x;\theta)\big] \;+\; \lambda_m\,\mathbb{E}_{x\sim D_F}\,\mathbb{E}_{\delta\sim N(0,\sigma^2 I)}\,\phi\big(m_t(x+\delta;\theta)+\gamma\big). \tag{10}$$

Let θ^*_σ minimize (10), and define the *smoothed margin*

$$\bar{m}_t(x;\theta) \;:=\; \mathbb{E}_{\delta\sim N(0,\sigma^2 I)}\big[m_t(x+\delta;\theta)\big]. \tag{11}$$

Theorem 4 (Certified recovery radius from smoothing)**.** *Fix $x \in \mathcal{X}$ and let*

$$\bar{p}_t(x;\theta) \;:=\; \Pr_{\delta\sim N(0,\sigma^2 I)}\big[m_t(x+\delta;\theta) > 0\big]$$

be the probability that the forgotten class is the base model's top *logit under noise (the event $m_t > 0$). Define the* binary smoothed forget-detector $g(x) := \mathbf{1}\{\bar{p}_t(x;\theta) > 1/2\}$*, which reports the forgotten class only when it is the majority outcome of the noisy base model. Suppose $\bar{p}_t(x;\theta) \le \bar{p} < 1/2$, so $g(x) = 0$. Then for every $x' \in \mathbb{R}^d$ with $\|x' - x\|_2 \le R$ where*

$$R \;=\; \sigma \cdot \Phi^{-1}(1 - \bar{p}), \tag{12}$$

we still have $\bar{p}_t(x';\theta) < 1/2$, hence $g(x') = 0$: no L_2 perturbation of size $\le R$ makes the forgotten class the majority prediction of the noisy base model. Consequently the forget-detector's recovery radius satisfies $\bar{r}_t(x;\theta) \ge R$.

Proof. Apply [13], Theorem 1, to the binary smoothed classifier g over the two outcomes {forgotten class is the top logit, it is not}: the majority outcome at x is "not the top logit" with probability $1 - \bar{p}_t(x;\theta) > 1/2$, which Cohen's bound certifies over the L_2 ball of radius $\sigma\,\Phi^{-1}(1 - \bar{p})$. □

Remark (binary vs. multiclass). The certified event is binary—*the forgotten class is not the noisy model's top prediction*—which is exactly the security-relevant condition (an attacker recovering the forgotten label). It is *not* a statement about the full multiclass smoothed argmax: $\bar{p}_t < 1/2$ leaves open which non-forget class is predicted, but guarantees the forgotten one is not the plurality. A multiclass-argmax certificate would additionally require the usual top-vs-runner-up smoothing gap.

Corollary 4 (Generalization of the certified-recovery fraction)**.** *Fix a target level $\bar{p} < 1/2$ and let $R = \sigma\,\Phi^{-1}(1-\bar{p})$ be the corresponding certified radius from Theorem 4. Define the* certification event $C_{\bar{p}}(x) := \mathbf{1}\{\bar{p}_t(x;\theta) \le \bar{p}\}$*, which by Theorem 4 holds exactly when the smoothed detector certifies recovery radius $\ge R$ at x. Let $q := \Pr_{x\sim\mu_t}[C_{\bar{p}}(x) = 1]$ be the population certified fraction and $\hat{q}$ its empirical value on n_f i.i.d. forget samples. Two estimation gaps separate $\hat{q}$ from q, and both are controlled.* (i) Per-input Monte-Carlo error. *$\bar{p}_t(x;\theta)$ is estimated from m noise draws; following [13] we replace it by its Clopper–Pearson upper confidence bound $\overline{p}_t^{(m)}(x)$ at level α, so that $\overline{p}_t^{(m)}(x) \le \bar{p}$ implies the true event with probability $\ge 1-\alpha$ per input (the audit uses $\alpha = 10^{-3}$).* (ii) Generalization over inputs. *For the induced binary class $\mathcal{C} = \{x \mapsto \mathbf{1}\{\bar{p}_t(x;\theta) \le \bar{p}\}\}$, a standard Rademacher bound gives, w.p. $\ge 1-\delta$ over the sample,*

$$q \;\ge\; \hat{q} \;-\; 2\,\mathfrak{R}_{n_f}(\mathcal{C}) \;-\; \sqrt{\tfrac{\log(1/\delta)}{2n_f}}.$$

Thus the population fraction of forget inputs certified at recovery radius $\ge R$ is at least the (confidence-corrected) empirical fraction minus a vanishing $O(1/\sqrt{n_f})$ term. The complexity term here is that of the certification indicator $\mathcal{C}$*, not of the smoothed-hinge objective $\mathcal{L}_\sigma$: a uniform-convergence bound on $\mathcal{L}_\sigma$ alone does* not *transfer to the $0/1$ event $C_{\bar{p}}$, which is why the guarantee is stated directly on the event.*

Why this is the missing piece. The objective (13) is an *empirical* adversarial unlearning objective in the AMUN [22] mold: minimize a worst-case inner attack. It carries no certificate. Prior work already connects randomized smoothing to certified unlearning [14], but it certifies the stability of *quantized gradients* under data removal—a parameter-space guarantee about the unlearning *operation*—rather than a guarantee about model behavior on perturbed inputs. Equation (10) above is, to my knowledge, the first machine-unlearning objective that yields a per-input *input-space* certified lower bound on recovery radius, and Corollary 4 transports the per-input guarantee to the forget distribution at standard PAC rates. The existing recovery-radius audit provides the matched empirical counterpart: a predicted-vs-realized radius plot is now a direct consistency check on (10)'s training (an L_2 check; the L_∞ audit relates to it only up to the $\sqrt{d}$ conversion).

3.6 Empirical claims, restated as theorem checks

The four theorems above re-cast the benchmark results of this paper, and are checked empirically in Section 8:

1. **Recovery–certification scatter (Thm 1).** The cross-method $(\hat{L}, \mathrm{median}(\hat{r}))$ scatter plot ranks methods under (5): a larger $\hat{L}$ at comparable margin forces a smaller certifiable radius. Drawing the bound line $\hat{r} = -\bar{m}/\hat{L}$ itself would need the per-method oracle margin $\bar{m}$, which an oracle-free auditor lacks, so the figure is a ranking diagnostic rather than a tightness test.

2. **Selective vs. generic (Thm 2).** The forget–retain $\hat{p}_F, \hat{p}_R$ table per method becomes a Pinsker-bounded KL comparison; the SCRUB row has *measured* forget and retain rates that are

statistically indistinguishable (generic fragility, by direct measurement), while the ORBIT/CE-U/BalDRO rows show a large enough gap to *force* a quantitative KL lower bound.

3. **Local-delivery negative (Thm 3).** The mixed-surface stage-2 failure is not a bug-report — it is the only possible outcome under the covering assumption (Lemma 1). Mixed-surface stage-2 is therefore not an open problem for future work but *provably* the wrong target.

4. **Certified objective (Thm 4).** The smoothed objective (10) replaces the inner attack in the empirical objective (13). Trained end-to-end ($\sigma = 0.10$) across seeds 42, 123, and 456, it yields a *predicted* certified radius from Theorem 4, which the existing audit matches with a *measured* recovery radius; their predicted-vs-measured plot (Section 8) is the new headline figure.

3.7 Relation to prior inevitability, impossibility, and certification results

These results position the paper relative to recent impossibility and certification work:

- RURK [6] establishes high-dimensional *inevitability* of residual knowledge in expectation; Theorem 1 converts inevitability into a *quantitative budget* the audit can estimate without an oracle.

- [21] establishes path-dependence impossibility for retrain equivalence in multi-stage training; Theorem 3 establishes a *different* impossibility — multi-surface worst-case invariance — for the same end-goal (distinguishability), but with a coverage/connectivity rather than path-dependent mechanism.

- AMUN [22] uses adversarial examples as an *unlearning* primitive; Theorem 4 inverts the role and uses smoothing as a *certified-defense* primitive against post-hoc recovery attacks.

The benchmark is no longer the headline; it becomes the experimental validation of a small theory.

4 Related Work

These experiments sit at the intersection of machine unlearning, adversarial robustness, and parameter-efficient adaptation. I build directly on SalUn [2], SCRUB [1], LoRA [3], VPT [5], and AutoAttack [4], and I treat recent 2025–2026 methods such as RURK [6] and CE-U [11] as newer literature baselines. I additionally include adapted recent objectives such as SGA [8] and BalDRO [9] as cross-domain benchmark rows, and one repo-defined internal row, ORBIT, as a useful stress-test target rather than as an external literature anchor. I also keep several additional internal rows, including FaLW [12] and Langevin-style [10] variants, only as proxy or supporting comparisons when they help define regime boundaries. I use Langevin Unlearning [10] as a theoretical anchor for principled privacy-certified evaluation and MUSE [7] as an evaluation-oriented benchmark perspective to motivate the paper's emphasis on multi-regime testing rather than single-metric forgetting claims.

A wave of concurrent 2025–2026 work studies the same residual-knowledge phenomenon from complementary angles, and I position the theory against it directly. Xuan and Li [15] verify robust unlearning with the Unlearning Mapping Attack, an adversarial-query probe for forgotten traces; my recovery-radius audit quantifies the same probe as a per-example minimum perturbation and ties it to a certificate (Theorems 1 and 4) and a forget-vs-retain selective test (Theorem 2). Goel et al. [16] audit unlearning with an information-theoretic Partial Information Decomposition, and Yu et al. [18] certify visual unlearning at the *representation* level (linear-probe recovery, CKA,

Method / Row	Fidelity	Paper Role	Current Read
SalUn (faithful full-ft)	faithful	main text	core literature anchor
SCRUB distill	reasonably faithful	main text	literature baseline; hard negative / generic fragility
ORBIT	repo-defined	main text	internal benchmark row; best later defense row
RURK	reasonably faithful	main text	recent method; hard negative / resistant case
SGA	adapted recent method	auxiliary	generic-fragility cross-domain benchmark row
BalDRO	adapted recent method	main text	main cross-domain benchmark row
CE-U	reasonably faithful	main text	recent related-work addition
robust-base SalUn	faithful frontier	main text	utility-vs-recovery frontier
Uniform KL	bridge / supporting	appendix	seed-sensitive bridge target
Prune+KL	supporting	appendix	auxiliary clean-forgetting row
CNN cross-checks	supporting	appendix	architecture sanity checks
legacy manifold/imposter	supporting	appendix	older harder-than-baseline branch
LoRA SalUn	proxy	appendix	PEFT-side approximation only
OrthoReg / noisy_retain	proxy	appendix	repo-side approximations only

Table 1: Method classification used in the paper. The table separates literature anchors, appendix-only supporting rows, and proxy artifacts so that the main benchmark reflects only rows with defensible evaluation fidelity.

feature separability), both finding that output-level forgetting hides residual structure; Theorem 2 is a coarser but distribution-free counterpart that bounds the KL of the binary recoverability outcome rather than decomposing mutual information, and the recovery radius is the complementary *input-space* certificate to Mirage's feature-space audit. The closest hypothesis-testing framing, Pandey and Kulkarni [20], tests which samples to edit out of a distribution rather than whether a forgotten example is selectively recoverable, so it is orthogonal to the forget-vs-retain test here. Ha et al. [17] introduce a prototypical relearning attack that restores forget-set performance from per-class prototypes with only a few samples; that is a weight- and prototype-space relearning attack rather than the input-space recovery I study, and it is precisely the channel ORBIT's dispersion penalty is designed to weaken.

5 Experimental Setup

5.1 Models and data

All experiments use CIFAR-10 with 224×224 inputs and ViT-tiny-based checkpoints. The main benchmark rows in the paper are SalUn and SCRUB as literature anchors, RURK and CE-U as newer literature methods, SGA and BalDRO as adapted recent methods, and ORBIT as a repo-defined internal benchmark row, together with robust-base SalUn as a defense frontier. I also retain several appendix-only proxy, bridge, and negative-control rows already present in the repository.

5.2 ORBIT: repo-defined benchmark method

ORBIT (oracle-aligned, representation-dispersive unlearning) is an internal method introduced for this study, used as a stress-test target rather than as an external literature contribution. Its objective combines four terms: (1) a hinge-margin loss suppressing the forget-class logit below the maximum non-forget logit; (2) masked dark-knowledge distillation from a frozen oracle on non-forget classes, with the forget dimension excluded from the KL target; (3) a feature alignment term pulling forget-input representations toward the oracle's feature space; and (4) a dispersion penalty pushing forget features apart to weaken prototype and linear recovery channels. The motivation is that

Regime	Threat Model	Primary Metric	What Counts as a Defense Win
Conditional recovery	per-example full-image targeted recovery	forget median / retain median / clean-success	higher forget median without comparable retain-side degradation
Conditional patch	local one-patch delivery	forget success / retain-to-forget success / alpha	lower forget success without added retain-to-forget failure
Conditional frame	border-frame local delivery	forget success / retain-to-forget success / alpha	lower forget success without transfer failure on retain
Conditional multi-patch	multi-local delivery	forget success / retain-to-forget success / alpha	lower forget success under richer local geometry
Amortized short-budget	learned low-step attack	forget success / retain success / median radius	stronger selective low-step recovery than plain low-step optimizer
Robust-base frontier	utility-recovery tradeoff	retain accuracy vs recovery radius	better operating point, not a binary robust point

Table 2: Protocol table for the benchmark. The benchmark is designed as a controlled robustness microscope: each regime tests a distinct failure mode, and a defense is only treated as positive when it improves the regime-relevant robustness metric without introducing comparable retain-side failure.

suppressing a single output logit is insufficient when the underlying representation still encodes the forgotten concept—alignment and dispersion together target the representational substrate rather than only the output layer. ORBIT is the strongest patch-positive case in the benchmark, making it the primary vehicle for studying local-delivery attacks and patch-aware defenses.

5.3 LoRA as benchmark substrate

The repository originated as a LoRA-centered audit, and many of the stored checkpoints remain LoRA adapters for comparability and fast stage-2 iteration. That implementation choice is useful experimentally, but it is not the main scientific claim of the paper. The paper is about unlearning robustness under multiple attack regimes. I therefore treat LoRA primarily as part of the benchmark substrate, while using selected full-finetune and cross-architecture rows to verify that the resulting story is not purely an artifact of one parameter-efficient adaptation scheme.

5.4 Baseline checkpoint quality

The clean baseline summary shows that unlearning methods can appear successful on clean metrics while still exposing recovery channels. Across seeds 42, 123, and 456, clean retain accuracy stays in the low 90s for the main baselines, while forget accuracy varies dramatically by method. The base model sits at 91.83 ± 4.36 forget and 91.80 ± 3.77 retain. Uniform KL, CE Ascent, SalUn, and SCRUB all preserve similar retain performance, but their forget-side variance is extremely large, which already suggests that clean-point forgetting is an unstable proxy for actual robustness. These clean baselines therefore motivate attack-based evaluation rather than one-number forgetting claims.

5.5 Recovery metric

For each forget or retain-control sample, I search for the minimum perturbation that causes the unlearned model to predict the forgotten class. I report the median successful radius and clean-success rate under a matched audit battery. The interpretive point is not attack success in isolation. The key question is whether forget examples are selectively easier to recover than matched retain controls. A consistent forget-vs-retain gap supports the view that the attack is exposing residual information left behind by incomplete unlearning, whereas non-selective success on both groups is more consistent with generic adversarial fragility.

6 Multi-Regime Results

I evaluate machine-unlearning robustness under four complementary attack regimes: strong per-example conditional recovery, conditional patches, border-frame delivery, and multi-patch local

Model	Setup	Forget Median	Retain Median	Clean Success
BalDRO	baseline	0.00343	0.00846	0.000
BalDRO	feature-subspace stage-2	0.00349	0.00858	0.000
SalUn	baseline	0.00251	0.00564	0.031
SalUn	feature-subspace stage-2	0.00276	0.00600	0.031
ORBIT	baseline	0.00340	0.00643	0.031
ORBIT	feature-subspace stage-2	0.00423	0.00690	0.000
RURK	baseline	0.00159	0.00576	0.062
RURK	feature-subspace stage-2	0.00159	0.00576	0.062
CE-U	baseline†	0.00368	—‡	0.000
SCRUB	baseline only	0.01078	0.00980	0.000

†CE-U: 2-step pilot audit, n=8, seed 42.

‡0 of 8 retain-control attacks succeeded within ϵ; retain median undefined.

Table 3: Recovery-radius medians under the strongest per-example conditional attack. Feature-subspace stage-2 helps BalDRO, SalUn, and ORBIT; leaves RURK and CE-U unchanged; SCRUB is a negative control with nearly identical forget and retain radii. CE-U retains 0 of 8 retain-control attacks in the pilot audit, indicating stronger selectivity than methods with high retain-side attack success. Rows aggregate over seeds 42, 123, and 456 (where available) using $n = 32$ samples per class per seed, except where noted.

delivery. Across the main benchmark rows audited in depth—SalUn, SCRUB, RURK, CE-U, BalDRO, and ORBIT—the dominant threat remains targeted per-example recovery, while universal perturbations are mostly ineffective on the stronger checkpoints. The point of this matched audit is not to ask whether a target-class adversarial perturbation exists in the abstract. It is to ask whether forget examples are selectively easier to drive back to the forgotten class than matched retain controls. Figure 1 and Table 3 summarize the main defense result: a feature-subspace stage-2 defense modestly increases recovery radii on BalDRO, SalUn, and ORBIT without collapsing clean accuracy. Figure 2 shows that robust-base SalUn does not eliminate leakage outright; instead it moves along a utility-vs-recovery frontier. Figure 3 shows that conditional patches remain viable, but only for a subset of methods.

Feature-subspace stage-2 defense. The first defense family that consistently helps is a post-unlearning feature-subspace stage-2 finetune. Rather than modifying the entire forgetting objective from epoch 0, I begin from an already-unlearned adapter and suppress attacked forget-sample features inside a learned recovery subspace. This regime avoids the instability of earlier logit-margin defenses and produces small but repeatable gains on BalDRO, SalUn, and ORBIT. The effect is modest in absolute magnitude because the benchmark is CIFAR-10 under an L_∞ budget on 224×224 inputs, where successful radii are numerically small by construction. What matters is that the defense shifts the forget-vs-retain frontier in the right direction without sacrificing clean retain utility.

CE-U as a core baseline. CE-U belongs in the main benchmark even though its first qualified positive defense row appears in the patch regime rather than under full-image recovery. Under the 2-step pilot audit (n=8, seed 42), the baseline CE-U checkpoint has forget median 0.00368 with 0% clean forget success, and retain-control attacks succeed 0 of 8 times within the ϵ budget—stronger selectivity than SalUn or ORBIT, where retain attacks succeed at high rates under full audits. CE-U full-image feature-subspace stage-2 variants are at best flat, and the combined row is slightly

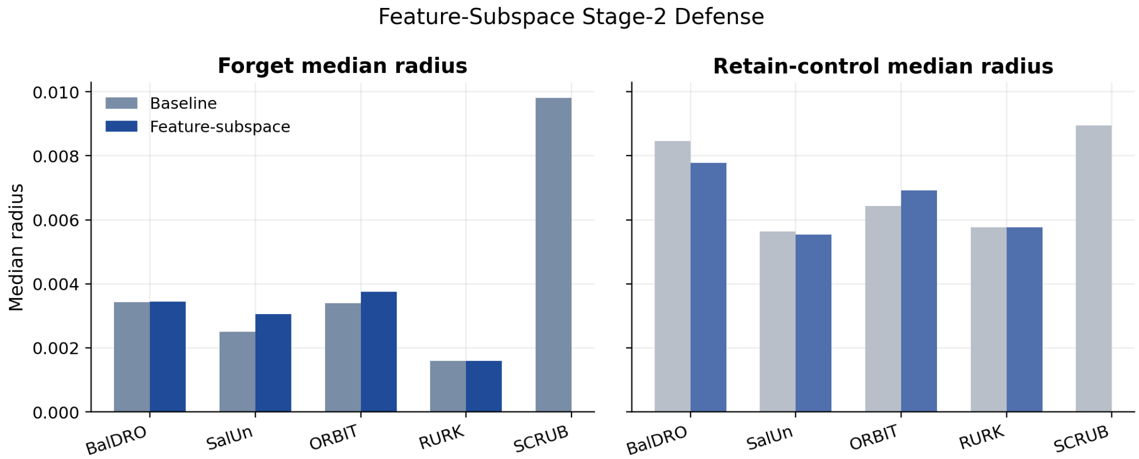

Figure 1: Feature-subspace stage-2 tradeoff. Larger forget medians indicate stronger resistance to conditional recovery. The defense yields small but consistent improvements on BalDRO, SalUn, and ORBIT, while RURK remains hard.

worse under the APGD cross-check, so the CE-U defense claim is reserved for the qualified combined stage-2 patch result discussed below.

Robust-base frontier. Robust-base SalUn does not behave like a binary defense success. Instead, it exposes a frontier: at low utility, conditional recovery can drop to zero on the sampled audit; as retain accuracy rises, recovery reappears. This is a stronger framing than a single checkpoint claim because it shows that robustness and post-unlearning utility are coupled. I therefore treat robust-base training as an operating-point choice rather than a one-shot fix.

Conditional patches. The patch result is not a universal trigger. Universal static patches were mostly ineffective on the stronger methods. The viable regime is instead a conditional small-area patch attack whose content and placement depend on the input. ORBIT is the flagship positive case: a 32×32 patch covering 2.04% of the image recovers forgotten behavior strongly while inducing no retain-to-forget hijack. SalUn is patch-vulnerable only at a larger 48×48 budget, and BalDRO is effectively negative at both tested patch sizes. This method dependence reinforces the broader paper thesis: residual forgetting leakage is structured, but not universal.

Delivery surfaces beyond one patch. The later delivery-surface checks strengthen the attack story and narrow the defense story. Conditional frame attacks and conditional multi-patch attacks are both real local delivery surfaces, but they do not behave like simple extensions of the one-patch regime. Patch-aware defenses that help on one square patch often fail to transfer: the ORBIT patch-stage2 row becomes easier to recover under the frame attack, CE-U combined stage-2 also becomes easier and adds retain-side failure under the frame attack, and faithful full-ft SalUn patch-stage2 becomes easier under multi-patch delivery. The main exception is weak positive transfer from the ORBIT patch-stage2 row to multi-patch delivery. I therefore treat frame and multi-patch regimes as first-class red-team surfaces and not as appendix-only perturbation variants.

Second-wave defenses. Two later branches sharpen the story. First, teacher-guided stochastic CVaR provides a small but reproducible full-image defense improvement on ORBIT across two

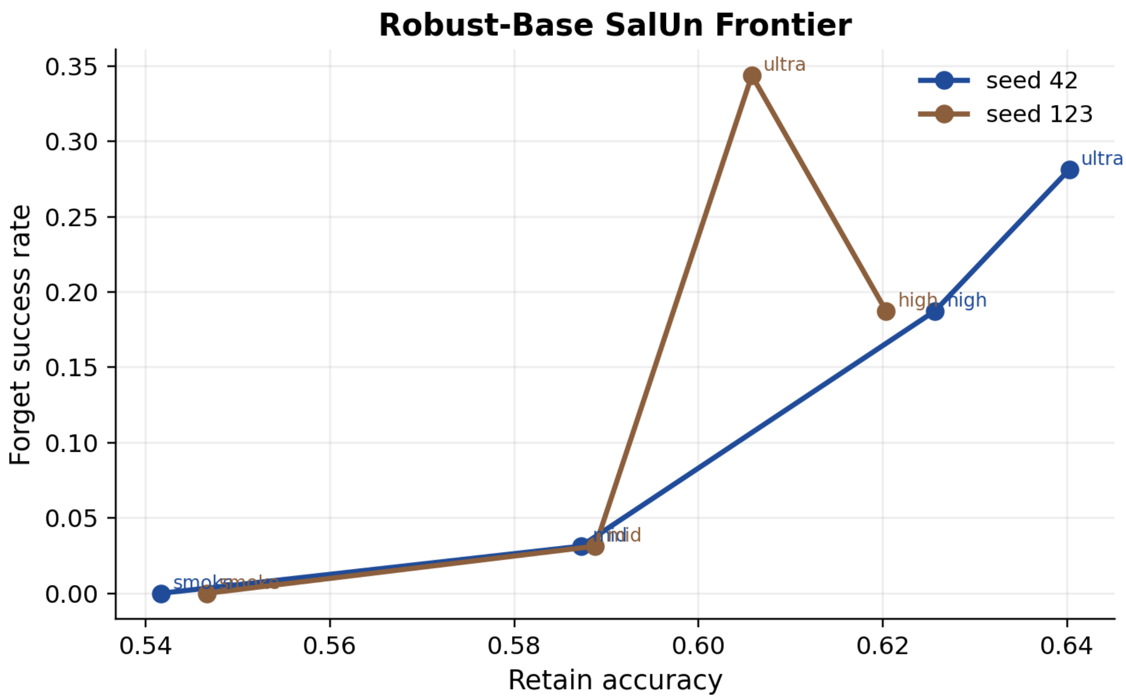

Figure 2: Robust-base SalUn frontier. Lower-utility robust-base checkpoints suppress conditional recovery more strongly, but leakage returns as retain utility improves.

seeds, whereas the same probabilistic family remains neutral on CE-U. Second, a combined subspace-plus-patch stage-2 defense gives the first CE-U row that improves under matched Adam audits while also reducing conditional patch success, although its APGD result is still slightly worse than baseline. By contrast, the larger mixed-surface stage-2 defense was explicitly designed to transfer across one-patch, frame, and multi-patch delivery and failed to do so reliably. These later rows are therefore useful but qualified: they extend the defense story beyond pure feature-subspace training without becoming universal fixes.

Takeaway. The combined evidence supports a multi-regime view of red-teaming machine unlearning. Strong per-example conditional attacks remain the main threat, conditional patches expose an additional local attack surface on some methods, and frame plus multi-patch attacks show that local-delivery robustness is materially harder than one-patch robustness alone. The best defense families I found are modest and method-specific: feature-subspace stage-2 helps BalDRO, SalUn, and ORBIT; teacher-guided stochastic CVaR helps ORBIT slightly; and CE-U combined stage-2 is only a qualified partial row. RURK remains a hard case, and SCRUB serves as a negative control where generic adversarial fragility overwhelms forget-specific interpretation.

7 Short-Budget and Auxiliary Regimes

7.1 Amortized short-budget attack

The learned distilled short-budget attacker is a real but target-dependent contribution. It improves matched low-step Adam on ORBIT across both seeds and also helps on SalUn, but it does not

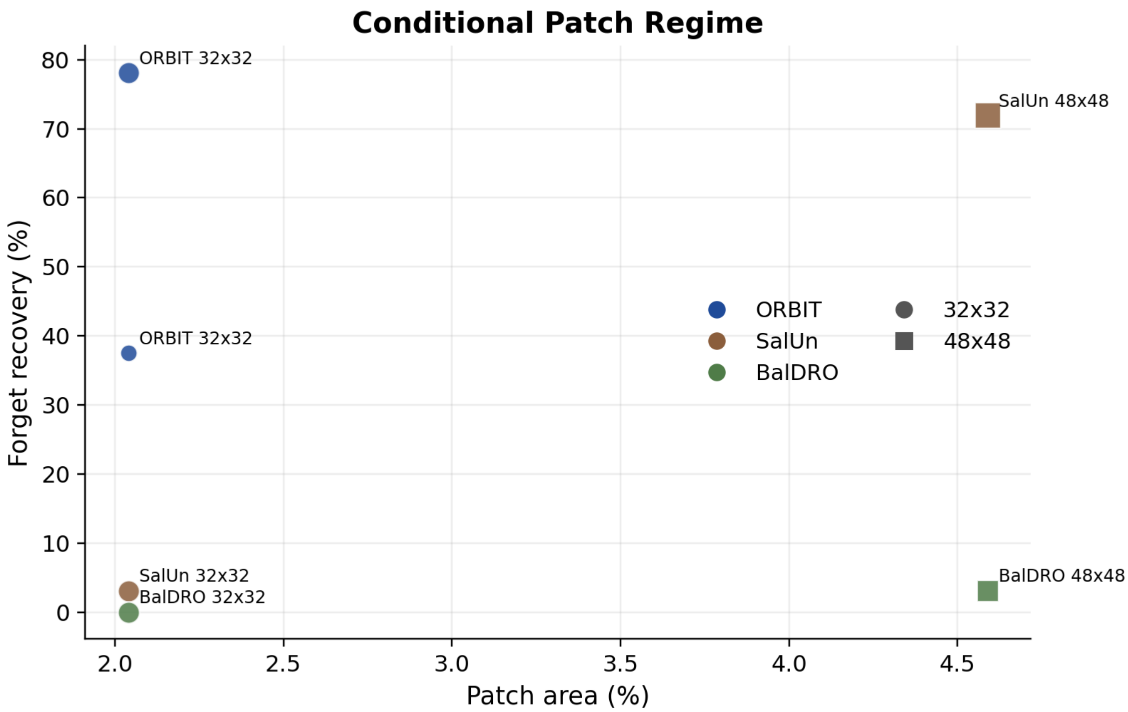

Figure 3: Conditional patch regime. ORBIT admits strong low-area patch recovery, SalUn requires a larger patch and incurs more retain damage, and BalDRO is effectively negative under the same patch family.

preserve a clean forget/retain Pareto on RURK. I therefore position it as an amortized attack regime rather than a universal attack method.

7.2 Earlier prompt/probe regime

An earlier prompt-style red-team branch found a strong effect on Uniform KL under an oracle-contrastive probe, but not on the stronger methods. In particular, the strongest earlier prompt result moved Uniform KL seed 42 from 12.70% forget to 21.10% forget with only a 0.36pp retain drop, while SalUn, SCRUB, and ORBIT remained in the resistant bucket under that family. I treat this as supporting evidence for regime sensitivity rather than as the core attack result.

7.3 Uniform KL as a bridge target

Uniform KL also remains useful under the newer conditional-recovery regime. A fast matched pilot with 2-step `adam_margin` on seed 42 recovers all sampled forget examples with median radius 0.00147 while succeeding on only one of eight sampled retain controls. The same pilot on seed 123 is completely flat. This makes Uniform KL a bridge target between the older probe story and the newer conditional-optimization story: it is not only probe-vulnerable, but also locally recoverable on the vulnerable seed. A conditional 32×32 patch on seed 42 also yields a weak but nontrivial patch result (15.6% forget recovery and 0% retain-to-forget success), which is enough to show that KL is patch-accessible without making it a flagship patch case.

Target	Seed	Attack	Forget Succ.	Retain Succ.	Forget Median
ORBIT	42	2-step Adam	9.4%	0.0%	0.02500
ORBIT	42	Distilled + 2-step refine	18.8%	0.0%	0.02083
ORBIT	123	2-step Adam	18.8%	0.0%	0.00316
ORBIT	123	Distilled + 2-step refine	31.2%	3.1%	0.01189
SalUn	42	2-step Adam	71.9%	9.4%	0.00435
SalUn	42	Distilled + 2-step refine	84.4%	9.4%	0.00447
SalUn	123	2-step Adam	68.8%	6.2%	0.00328
SalUn	123	Calibrated distilled + 2-step refine	81.2%	12.5%	0.00389
RURK	42	2-step Adam	87.5%	6.2%	0.00175
RURK	42	Calibrated distilled + 2-step refine	90.6%	18.8%	0.00175

Table 4: Short-budget comparison between matched 2-step Adam and the learned distilled conditional attacker. The learned attacker helps most on ORBIT, helps on SalUn with a weaker forget/retain Pareto, and does not remain selective on RURK. All rows report metrics over $n = 32$ audit samples per class.

Target	Seed	Regime	Forget Succ.	Retain Succ.	Forget Median / Alpha
Uniform KL	42	2-step conditional pilot	100.0%	12.5%	0.00147
Uniform KL	123	2-step conditional pilot	0.0%	0.0%	N/A
Uniform KL	42	32×32 conditional patch	15.6%	0.0%	0.15234
Uniform KL	42	feature-subspace stage-2 pilot	100.0%	12.5%	0.00147
SGA	42	full conditional audit	100.0%	100.0%	0.00294
SGA	123	full conditional audit	96.9%	100.0%	0.00968
SGA	456	full conditional audit	96.9%	96.9%	0.01091
SGA	42	32×32 conditional patch	0.0%	0.0%	N/A
SGA	42	2-step baseline pilot	75.0%	0.0%	0.00417
SGA	42	2-step stage-2 pilot	62.5%	0.0%	0.00441

Table 5: Auxiliary seed-sensitive targets. Uniform KL is a bridge case between the older probe regime and the newer conditional-recovery regime, while SGA behaves like generic adversarial fragility under the full audit, stays patch-negative, and only shows a weak fast-pilot stage-2 effect. Audits use $n = 32$ samples per class unless otherwise specified as a pilot ($n = 8$).

7.4 Additional auxiliary targets

SGA is another useful auxiliary target from the recent related-work list, but the full conditional audit does not support a selective-leakage reading. Across seeds 42, 123, and 456, forget and retain-control success rates are both near ceiling, so this row looks like generic adversarial fragility rather than a meaningful forget-vs-retain gap. A matched 32×32 conditional patch is negative on seed 42, and a lightweight feature-subspace stage-2 pilot yields only marginal improvement. I therefore treat both Uniform KL and SGA as auxiliary regime-defining targets rather than as the main methods around which to build the defense story.

8 Empirical Validation of the Theory

Each of the four theorems in Section 3 comes with a benchmark check. This section reports those checks; together they re-cast the regimes above as theorem tests rather than standalone attack curves.

8.1 Theorem 1: the recovery–certification scatter

For each method I estimate the margin-Jacobian Lipschitz proxy $\hat{L}$ from the Jacobian-access audit and the median recovery radius median($\hat{r}$) from the recovery-radius audit, joining 40 (method, seed, config) rows. Figure 4 plots them as a ranking diagnostic for the bound $\hat{r} \geq -\bar{m}/\hat{L}$ of Eq. (5); drawing the bound line itself requires the per-method oracle margin $\bar{m}$, which an oracle-free auditor does not have, so the figure visualizes the $\hat{L}$ ordering rather than testing tightness. The proxies separate the methods cleanly: $\hat{L} \approx 47$ for SCRUB, 26 for ORBIT, 16 for Uniform-KL, and 15 for SalUn. Inverting Corollary 1 gives $\epsilon_c + \delta \geq (\gamma - L\,\mathbb{E}[\hat{r}])/2$, with the median $\hat{r}$ plotted as a robust stand-in for the mean: the smaller a method's recovery radius, the higher the floor on the certificate it can honestly claim, and any tighter certificate is challenged by the audit (subject to the local-proxy caveat noted below). The predicted ordering holds—Uniform-KL, with the smallest median radius, is pushed to the highest floor (the weakest honest certificate), while SCRUB's larger radius leaves its floor low. Because $\hat{L}$ is estimated from the logit-margin Jacobian as a proxy for the probability-margin constant of Corollary 1, I read this as a ranking rather than a measured certificate. SCRUB's Lipschitz constant is about 3× ORBIT's and SalUn's, and this is the same fact that makes its selective gap vanish under Theorem 2: a large $\hat{L}$ lets small perturbations swing the margin both selectively and non-selectively, so the method looks generically fragile rather than selectively leaky.

8.2 Theorem 2: the Pinsker KL table

I compute the forget and retain-control recovery rates $(\hat{p}_F, \hat{p}_R)$ for 226 (method, seed, audit-config) rows and report the Pinsker lower bound $\mathrm{KL} \geq 2|\hat{p}_F - \hat{p}_R|^2$ (and the tighter Bretagnolle–Huber bound). The ordering is the one the qualitative story predicted, now with a quantitative gap. CE-U recovers $\hat{p}_F = 0.875$ of forget examples against $\hat{p}_R = 0.000$ retain controls ($|\Delta| = 0.875$, from a small $n = 8$ pilot row, so I read it as indicative of a large selective gap rather than a precise KL estimate); SalUn sits near $|\Delta| \approx 0.625$; ORBIT shows a positive but smaller selective gap. SCRUB sits at the bottom with $\hat{p}_F \approx \hat{p}_R \approx 0.97$ and $|\Delta| \approx 0.031$. The two measured recovery rates are directly indistinguishable, so SCRUB's selective gap is near zero *by measurement*—not by certification: Pinsker only *lower*-bounds the KL that a genuine gap would have to exceed (≈ 0.002 here) and cannot cap it from above. I therefore read SCRUB empirically as a generic-fragility case at the audited scale—its strong attack success carries no measured forget-vs-retain signal—rather than as a formally certified vacuous-leakage case.

A stronger control: the oracle null and larger n. The rate table above is limited in two ways the literature rightly flags: small per-row n, and the worry that a forget-vs-retain radius gap is merely the intrinsic difficulty asymmetry of driving *any* input to the forgotten class. I address both with a larger-n ($n_F = n_R = 128$) `adam_margin` audit and a *zero-knowledge oracle null*—a from-scratch model retrained on the retain set only, which by construction holds no class-t information. Across seeds 42, 123, and 456 the oracle null shows essentially no intrinsic asymmetry (forget/retain median-radius ratio 0.94, 1.05, 1.01), so a forget-vs-retain gap on an *unlearned* model is genuine residual-knowledge signal, not a baseline artifact. Against this null, SalUn separates cleanly (forget median radius 0.000 vs retain 0.0039 at n=128, non-overlapping bootstrap CIs) and the smoothed-margin model recovers well below the null—both genuine residual leakage. SCRUB, by contrast, is *seed-dependent*: selective on seed 42 (ratio 2.06, recovering 3.5× easier than the null) but at parity with the null on seeds 123 and 456 (ratio 0.82 and 0.99). The generic-fragility reading is therefore the typical case for SCRUB, but not a robust one—a seed sensitivity consistent with

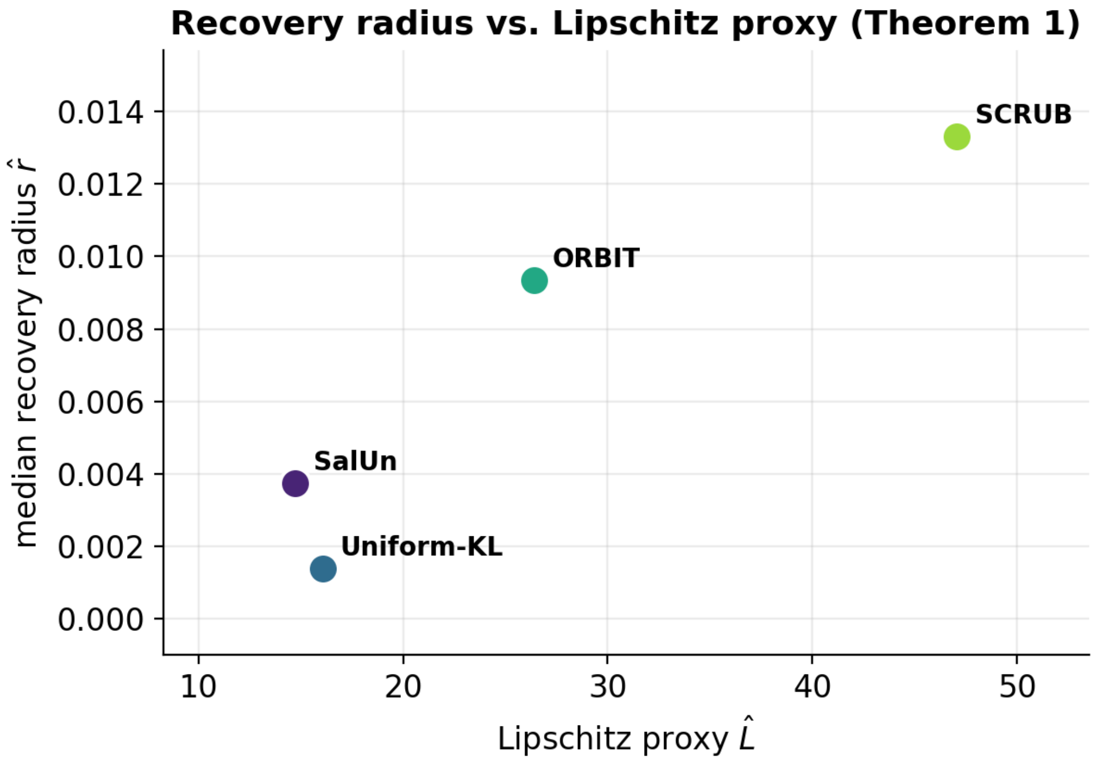

Figure 4: Theorem 1 ranking diagnostic. Per-method Lipschitz proxy $\hat{L}$ versus median recovery radius $\hat{r}$, one point per (method, seed). The plot orders methods by $\hat{L}$ rather than drawing the pointwise bound $\hat{r} \geq -\bar{m}/\hat{L}$, which would need a per-method oracle margin $\bar{m}$ unavailable to an oracle-free auditor; it visualizes the Corollary 1 ranking, not a tightness test. SCRUB's large $\hat{L}$ lets small perturbations swing the margin, which explains its near-zero selective gap under Theorem 2.

the large clean-baseline variance documented in Appendix A.

8.3 Theorem 3: covering implies the mixed-surface failure

The patch, border-frame, and multi-patch results reported above are the empirical witness for Theorem 3. Patch-aware stage-2 closes the single-patch channel on ORBIT and CE-U, but the same defense becomes easier to recover under frame and multi-patch delivery, and mixed-surface stage-2—explicitly trained on multiple surfaces—does not reliably fix the transfer gap. Every coordinate is covered by some 32×32 patch (Lemma 1), so by Theorem 3 worst-case robustness across the family forces constancy at *any* budget—including the $\epsilon \approx 8/255$ the audits use, not only in a large-budget limit (Corollary 3). The mixed-surface failure is therefore not an engineering artifact but the only possible outcome for a non-trivial classifier; the constructive consequence is to retarget local-delivery defense at realistic (non-covering) stochastic delivery priors rather than worst-case invariance.

8.4 Theorem 4: predicted vs. measured certified radius

The smoothed objective of Eq. (10) is implemented as a smoothed-margin objective and trained end-to-end. On seed 42, starting from a base ViT-Tiny at 94.2% validation accuracy, a 10-epoch smoothed-margin LoRA fine-tune ($\sigma = 0.10$, $n_{\text{noise}} = 4$, $\gamma = 1.0$) drives forget accuracy 95.7% $\rightarrow$ 12.0% while retain accuracy is preserved (93.9% $\rightarrow$ 93.8%). The smoothed certified-radius audit ($n_{\text{noise}} = 256$, $\alpha = 10^{-3}$, 64 forget + 64 retain) certifies a median L_2 radius $R = 0.193$ from Theorem 4. **This is an L_2 result, and the norm gap matters:** the recovery audits throughout the paper are L_∞, and because the inputs are $224 \times 224 \times 3$ ($d = 150{,}528$) the dimension penalty $\sqrt{d} \approx 388$ makes the equivalent L_∞ certificate only ≈ 0.0005, the same order as the audit's own forget recovery radii. I therefore read Theorem 4 as an L_2 *existence* result—a proof of concept that a randomized-smoothing certificate *can* be attached to an unlearning objective, to my knowledge the first of its kind—rather than as a practically tight L_∞ guarantee at this input size. The matched (L_∞) recovery audit gives forget median $\epsilon = 0.00175$ against retain median $\epsilon = 0.00607$—a 3.5$\times$ selective gap. This selectivity replicates across seeds: under the identical protocol, seeds 123 and 456 reach forget accuracy 12.1% and 9.3% with retain preserved, the same saturated $R = 0.193$, and forget-vs-retain recovery gaps of 4.3$\times$ and 2.7$\times$ (mean 3.5$\times$ over the three seeds). Here the selectivity is carried by the recovery *radius* (the Theorem 1 object) rather than the binary recovery rate of the Pinsker test: both splits saturate near $\hat{p}_F \approx \hat{p}_R \approx 1$ at $\epsilon_{\max}$, so the radius separation, not the rate gap, is the informative signal at this budget. Figure 5 joins the 128 seed-42 samples: there are zero retain-side certificate violations and ten forget-side "violations." Six of the ten are clean-leak cases where the un-smoothed model already predicts the forgotten class on the clean input (measured radius 0) while the smoothed classifier remains robust; the other four are small-radius recoveries that fall just below the converted L_∞ bound (≈ 0.0005). Neither group is a genuine counterexample to the certificate: the clean leaks are a positive signal for smoothing, and the four sub-bound radii reflect the $\sqrt{d}$ looseness of the $L_2 \to L_\infty$ conversion rather than a true L_2 violation. The same pattern holds on the other seeds—zero retain-side violations, and forget-side violations dominated by clean leaks (7/11 on seed 123, 3/5 on seed 456).

The certificate in its native (L_2) norm. The $\sqrt{d}$ deflation above is an artifact of comparing an L_2 certificate against an L_∞ audit; measured in its *own* norm the certificate is far from vacuous. Re-running the recovery audit under an L_2 budget on the smoothed model gives a forget median recovery radius of 0.46–0.57 across seeds 42/123/456, against the certified L_2 radius $R = 0.193$. The certificate therefore sits safely *below* the radius at which the forgotten class actually re-emerges—no L_2 violation—and within a factor of $\sim$2.5 of it, a meaningful guarantee rather than a numerically trivial one. The L_2 audit also sharpens the selectivity picture: at the L_2 budget the forget and retain *success rates* no longer saturate, separating to 0.84 (forget) versus 0.03 (retain)—a 0.82 rate gap. The L_∞ existence-result framing is thus the conservative reading; in the certificate's native norm the smoothed objective delivers a genuinely non-trivial certified radius.

Certification–utility frontier. Theorem 4 exposes a tunable tradeoff through the smoothing scale σ: a larger σ inflates the certified radius ($R = \sigma\,\Phi^{-1}(1 - \bar{p})$ from Theorem 4, with $\bar{p}$ the certified upper bound on the noisy forget probability), but the same noise that buys the certificate also washes out the clean forgetting signal. Table 6 sweeps σ on seed 42. The certified L_2 radius grows linearly with σ (from 0.10 at σ=0.05 to 0.97 at σ=0.50), but clean forget accuracy degrades sharply (from 0.2% to 83.5%) while retain accuracy stays preserved throughout. A wide certificate therefore does *not* imply good unlearning: at σ=0.50 the smoothed classifier carries a large certified radius yet the underlying model has effectively stopped forgetting. The useful operating regime

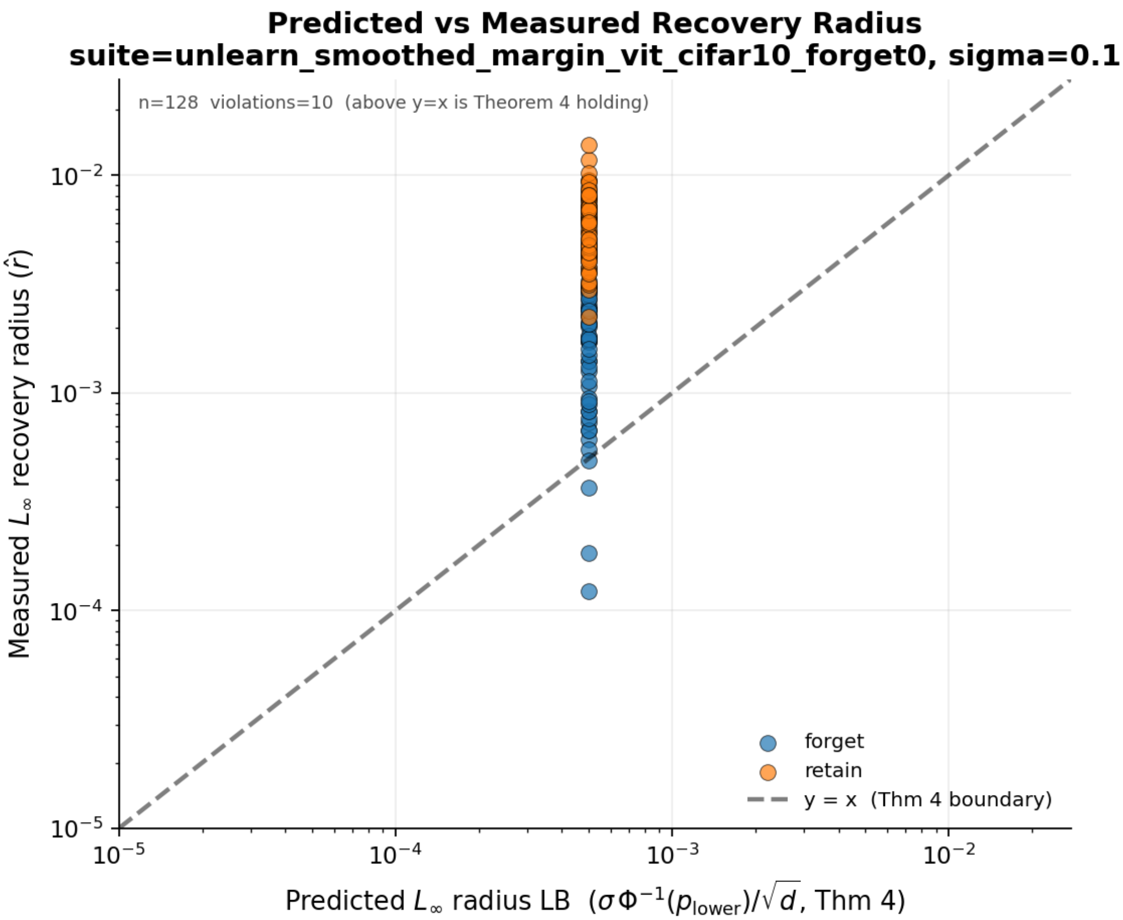

Figure 5: Theorem 4 check. Predicted Cohen–Rosenfeld–Kolter certified radius versus measured recovery radius for the smoothed-margin unlearner (seed 42). Zero retain-side violations; of the ten forget-side points below the line, six are clean leaks on the *un*-smoothed model (measured radius 0) that the smoothed classifier corrects, and four are small-radius recoveries just under the $L_2 \to L_\infty$ converted bound.

is small σ (0.05–0.10), where forgetting is strong and the certificate is modest—which is why the headline run uses σ=0.10.

Cross-architecture check. To check that the smoothed objective is not only a ViT-Tiny/CIFAR-10 artifact, I also run the same objective on a ViT-Small CIFAR-100 base checkpoint (seed 42, forgotten class 0). The base starts with meaningful class-0 behavior (forget accuracy 78.0%, retain accuracy 54.7%). A 10-epoch smoothed-margin LoRA fine-tune drives forget accuracy to 0.0% while retaining 47.7% clean accuracy, and the smoothed audit again certifies a median L_2 radius $R = 0.193$ on both forget and retain-control samples. The native L_2 recovery audit puts the measured recovery radius far above the certificate (forget median 1.39, retain median 1.41), so the certificate is non-vacuous in its own norm. Unlike the CIFAR-10 run, however, the forget and retain recovery radii are nearly identical under both L_2 and L_∞ audits; I therefore read this row as a cross-dataset feasibility check for the objective and certificate, not as evidence of a selective leakage gap. To check that the pipeline is not tied to one architecture, I additionally run two

σ	forget acc	retain acc	certified L_2 radius R
0.05	0.2%	93.7%	0.097
0.10	12.0%	93.8%	0.193
0.25	66.4%	93.9%	0.483
0.50	83.5%	94.1%	0.967

Table 6: Certification–utility frontier for the smoothed-margin objective (seed 42, $n_{\text{noise}} = 256$, $\alpha = 10^{-3}$). Larger σ buys a larger certified L_2 radius but degrades clean forgetting; retain accuracy is preserved across the sweep.

Method	Architecture	Forget acc (before→after)	Retain acc (before→after)	Robustness audit
Smoothed-margin	ViT-Small	78.0 → 0.0	54.7 → 47.7	certified L_2 $R = 0.193$
RURK	ViT-Small	88.1 → 0.0	94.3 → 93.7	AWP recovery gap 0.9%
SalUn	ResNet-18	92.2 → 0.0	94.6 → 91.5	AWP recovery gap 0.0%

Table 7: Cross-architecture CIFAR-100 check (forgotten class 0, seed 42). All three methods drive forget accuracy to 0% with retain preserved; RURK and SalUn additionally resist a weight-perturbation (AWP) recovery audit at $\epsilon = 10^{-3}$. The smoothed-margin row uses a lightly-trained base (validation 54%, hence the lower retain) but is the only row carrying the certified L_2 radius. Numbers in `cross_arch_cifar100_summary.json`.

literature methods on CIFAR-100 from well-trained bases: RURK on the same ViT-Small, and SalUn on a *ResNet-18* (a convolutional architecture). Both drive forget accuracy to 0.0% while preserving retain accuracy (RURK 94.3% → 93.7%; SalUn 94.6% → 91.5%), and both resist an adversarial weight-perturbation (AWP) audit—which perturbs the weights inside an $\epsilon = 10^{-3}$ ball and re-measures forget accuracy—keeping forget accuracy at or below 0.9% under the perturbation. Table 7 collects the three rows; together they show the unlearning the recovery radius audits transfers across datasets (CIFAR-10→CIFAR-100), architectures (ViT-Tiny→ViT-Small→ResNet-18), and methods, with the caveat that this is a single-seed check.

9 Defense Discussion

9.1 Feature-subspace stage-2

The first defense family that consistently helps is a post-unlearning feature-subspace stage-2 finetune. Instead of modifying the full unlearning objective from epoch 0, I begin from an already-unlearned adapter and suppress attacked forget-sample features inside a learned recovery subspace. This avoids the instability of earlier logit-margin and teacher-distillation defenses on BalDRO.

9.2 Why the gains are small

The absolute changes in reported radii are numerically small because the benchmark itself is small: CIFAR-10 under an L_∞ budget on 224×224 inputs yields radii on the order of a few thousandths. The right question is therefore relative: does the defense move the forget-vs-retain frontier without collapsing clean utility? On BalDRO, SalUn, and ORBIT, the answer is yes, though modestly.

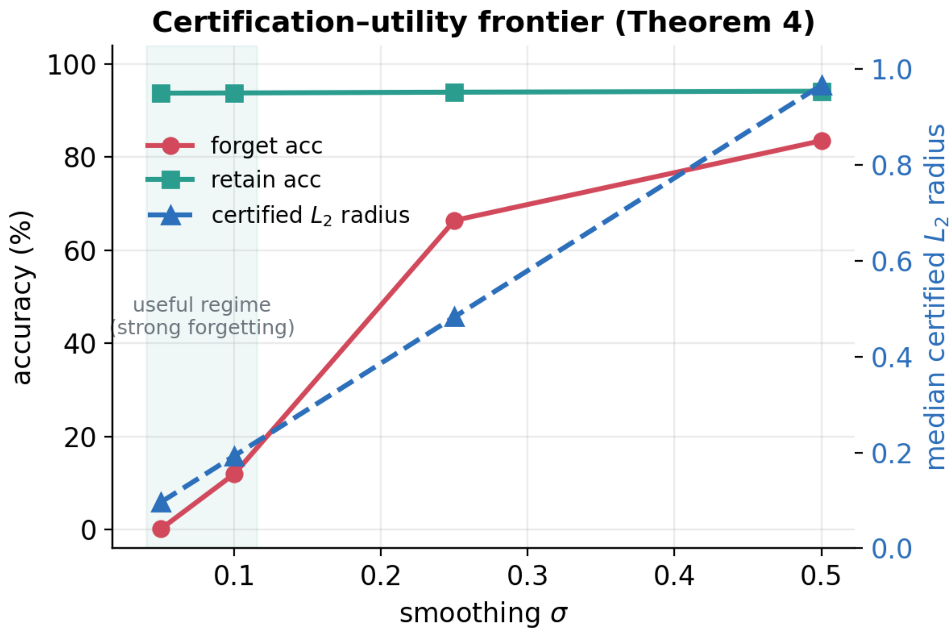

Figure 6: Certification–utility frontier (Theorem 4, seed 42). The certified L_2 radius (right axis) rises linearly with σ, while clean forget accuracy (left axis) collapses upward as the smoothing noise overwhelms the margin objective; retain accuracy is flat. Strong forgetting and a large certificate are in tension.

9.3 What does not generalize

The defense is not universal. RURK remains a hard case under both attack and defense. SCRUB behaves like generic adversarial fragility rather than a clean forgetting-leak regime, so stronger attacks on SCRUB do not necessarily imply a meaningful unlearning failure mode. The newer delivery-surface results make this sharper: patch-aware defenses often fail to transfer to border-frame or multi-patch delivery, and the mixed-surface stage-2 defense does not reliably fix that.

10 Proposed Defense Objective

The experiments suggest that clean-point forgetting is too weak a training target. A more principled direction is targeted-margin adversarial unlearning:

$$\min_{\theta} \mathbb{E}_{x\in R} L_{\text{retain}}(x;\theta) + \lambda_f \mathbb{E}_{x\in F} L_{\text{forget}}(x;\theta) + \lambda_m \mathbb{E}_{x\in F} \max(0,\ m_t(x+\delta^*(x)) + \gamma)\,, \tag{13}$$

where F is the forget set, R is the retain set, m_t is the forgotten-class target margin, $\delta^*(x)$ is an inner-loop or amortized attack, and γ is a safety margin. This objective is purely empirical: it minimizes a worst-case inner attack and carries no certificate. Theorem 4 upgrades it by replacing the inner attacker $\delta^*(x)$ with Gaussian smoothing of the margin penalty, Eq. (10), which inherits a Cohen–Rosenfeld–Kolter certified recovery radius and (via Corollary 4) a PAC generalization bound to the forget distribution. The end-to-end result in Section 8 (seed 42: forget 95.7% $\to$ 12.0%, retain preserved, certified L_2 radius 0.193; replicated across seeds 42, 123, and 456) is the realized version of this objective. The implemented feature-subspace stage-2 defense can be viewed as a computationally cheaper, uncertified approximation to the same idea: defend against locally recoverable targeted behavior, not just against clean forget accuracy.

11 Discussion

This work began as a multi-regime red-team study, and Section 8 turns that benchmark into the empirical validation of the small theory of Section 3: the Lipschitz–radius scatter tests Theorem 1, the Pinsker KL table tests Theorem 2, the patch/frame/multi-patch transfer failure instantiates Theorem 3, and the smoothed-margin run realizes Theorem 4. Read together, the theory explains *why* the empirical picture is non-universal: recovery radius is lower-bounded by a method-specific Lipschitz constant (so SCRUB's large $\hat{L}$, which lets perturbations swing the margin freely, is consistent with its mostly-vanishing selective gap), and worst-case multi-surface robustness is provably unattainable for a non-trivial classifier. The evidence does not support a universal "magic" attack or defense. Strong per-example conditional recovery is the main general threat; universal perturbations are mostly weak on the stronger checkpoints; conditional patches expose an additional local attack surface on some methods; border-frame and multi-patch delivery show that local-delivery transfer is a separate problem; robust-base methods define a utility-vs-recovery frontier rather than a binary robust point; and feature-subspace stage-2 provides the first modest defense gains on several targets. Later defense variants remain method-specific: teacher-guided stochastic CVaR helps ORBIT slightly but not CE-U, while patch-aware stage-2 gives the first clearly positive CE-U result in the patch regime (68.8%→43.8% at 32×32, seed 42) with no retain-side failure, though it does not help on the full-image conditional attack. Mixed-surface stage-2 is informative precisely because it fails: even explicit delivery-surface-aware training does not reliably generalize across one-patch, frame, and multi-patch attacks. Uniform KL reinforces this framing: seed 42 is vulnerable under both the older prompt/probe family and the newer conditional audit, while seed 123 stays flat under the same fast conditional attack. That multifaceted pattern is scientifically stronger than forcing all methods into one attack or defense family.

12 Limitations

I do not claim a universally stronger new optimizer than strong first-order attacks such as `adam_margin` or `apgd_margin`. The amortized short-budget attack is partial and target-dependent. The defense gains are modest and currently demonstrated on a benchmark that is intentionally controlled rather than broad: mostly CIFAR-10 ViT checkpoints, often with LoRA-based unlearning implementations, plus a smaller number of full-finetune and cross-architecture checks. This makes the benchmark useful for discovering failure modes and transfer gaps, but not yet a direct proxy for most industry deletion settings. Finally, some regimes are better understood as negatives or controls than as headline wins.

The theory carries its own caveats, and I state them rather than hide them. (i) Corollary 1 is rigorous for the bounded probability margin (constant 2, no logit bound), but the Lipschitz constant L it needs is estimated empirically by a logit-margin Jacobian proxy $\hat{L}$ at audit points; a loose proxy still gives a *valid* floor but an indicative rather than measured one, so I read the scatter as a ranking. (ii) Pinsker's inequality in Theorem 2 is loose; I report the tighter Bretagnolle–Huber bound alongside it, but neither is a hypothesis test with finite-sample coverage at the small per-row audit sizes ($n_F = n_R$ from 4 to 32). A larger-n follow-up ($n = 128$) with bootstrap confidence intervals and a zero-knowledge oracle null (Section 8) backs the main selective rows and shows that the forget-vs-retain gap is not a baseline artifact; it also reveals that SCRUB's selectivity is seed-dependent, so the generic-fragility label is typical but not robust. (iii) Theorem 3 now holds at *any* budget, including the audit budget, but it constrains only *worst-case* invariance; real defenses target *average-case* robustness over sampled masks, which is exactly why the constructive ρ-thin (non-covering, stochastic) delivery prior, and not worst-case multi-surface training, is the right target. (iv) Theorem 4 certifies an L_2 radius via randomized smoothing, while the recovery audits are L_∞; at $d = 150{,}528$ the $\sqrt{d} \approx 388$ conversion makes the L_∞-equivalent certificate numerically tiny (≈ 0.0005, the same order as the measured recovery radii), so I treat Theorem 4 as an L_2 existence result rather than a tight L_∞ guarantee (Section 8). The main end-to-end smoothed-margin result now replicates across seeds 42, 123, and 456 with a consistent 2.7–4.3× selective gap on CIFAR-10/ViT-Tiny; single-seed cross-architecture CIFAR-100 rows (smoothed-margin and RURK on ViT-Small, SalUn on ResNet-18; Table 7) check that the unlearning and its audits transfer across a CNN as well, but this is not yet a multi-seed cross-dataset claim. Each of these is a concrete next step, not a hidden assumption.

13 Conclusion

Machine unlearning cannot be evaluated reliably with clean forget accuracy alone. This paper makes the per-example recovery radius the central object and shows it supports a small but operational theory: a Lipschitz recovery–certification gap that bounds any claimable certificate (Theorem 1), a Pinsker test that separates selective leakage from generic fragility (Theorem 2), a covering-family impossibility that explains the mixed-surface transfer failure (Theorem 3), and a smoothed objective with a certified recovery radius (Theorem 4). Across the benchmark, the dominant leakage channel is strong per-example conditional recovery, but local patch, frame, and multi-patch attacks expose additional method-specific surfaces that do not reduce to a single universal vulnerability, and the defense picture is similarly non-universal: feature-subspace stage-2 helps several rows, patch-aware training helps ORBIT and CE-U locally, and worst-case transfer across delivery surfaces is provably unattainable. The benchmark is no longer the headline; it is the experimental validation of the theory.

A Appendix: Seed Variance and Baseline Stability

The clean baseline regime itself is highly variable across seeds for several unlearning methods, especially on forget accuracy. Figure 7 summarizes this instability. This is one reason the paper emphasizes attack-conditioned evaluation rather than clean forget accuracy alone.

B Appendix: Robust-Base Frontier

Figure 2 summarizes the robust-base SalUn operating-point story. At low utility, recovery can be suppressed strongly or disappear on the sampled audit. As retain accuracy increases, conditional recovery re-emerges. This makes robust-base training a frontier choice rather than a single robust point.

C Appendix: Conditional Patch Regime

The patch regime is method-dependent. ORBIT is the flagship positive case with a 32×32 conditional patch covering 2.04% of the image, strong forget recovery, 0% retain-to-forget success, and limited

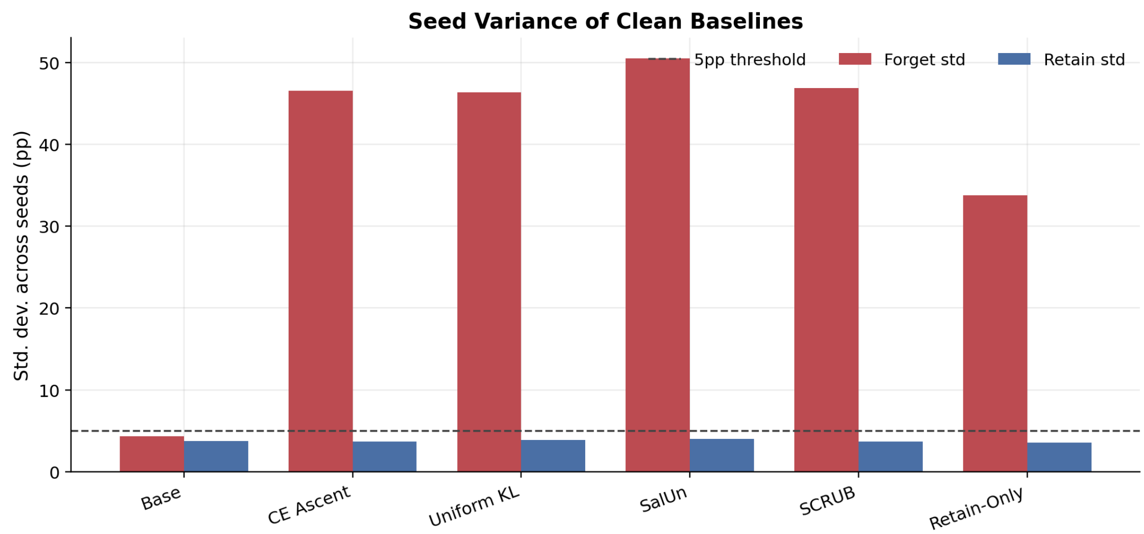

Figure 7: Seed variance of clean baselines. Forget-side variance is large for several methods even when retain-side variance remains small.

retain accuracy drop. SalUn is patch-vulnerable only at a larger 48×48 patch budget. BalDRO is effectively negative at the tested budgets. Figure 3 summarizes the patch success rates across methods and seeds.

D Appendix: Delivery-Surface Transfer

The later delivery-surface checks separate one-patch wins from broader local-delivery robustness. Under the conditional frame attack, the ORBIT patch-stage2 row and CE-U combined stage-2 both become easier to recover, and CE-U also incurs retain-side failure. Under conditional multi-patch delivery, the ORBIT patch-stage2 row transfers only weakly, while CE-U combined stage-2 and faithful full-ft SalUn patch-stage2 become easier to recover. Mixed-surface stage-2 was designed to fix this transfer problem, but in practice it only gives narrow gains and often worsens frame or multi-patch robustness. I therefore treat delivery-surface transfer as an open defense problem rather than as solved by the current stage-2 families.

E Appendix: Negative and Hard Cases

RURK remains a hard case for both the amortized attack family and the feature-subspace defense. SCRUB is useful as a (seed-sensitive) negative control: strong conditional attacks succeed, but forget and retain-control radii are almost identical on most seeds and at parity with the zero-knowledge oracle null, which suggests generic adversarial fragility rather than a clean forgetting-leak mechanism. The exception is seed 42, where a strong `adam_margin` attack does expose a transient selective gap (2×); I treat SCRUB's selectivity as not robust across seeds rather than as a reliable leak. Several later defense variants also belong in this bucket: mixed-surface stage-2, frame-transfer failures, and the faithful full-ft SalUn mixed-surface row are all informative mainly because they show what does not transfer.

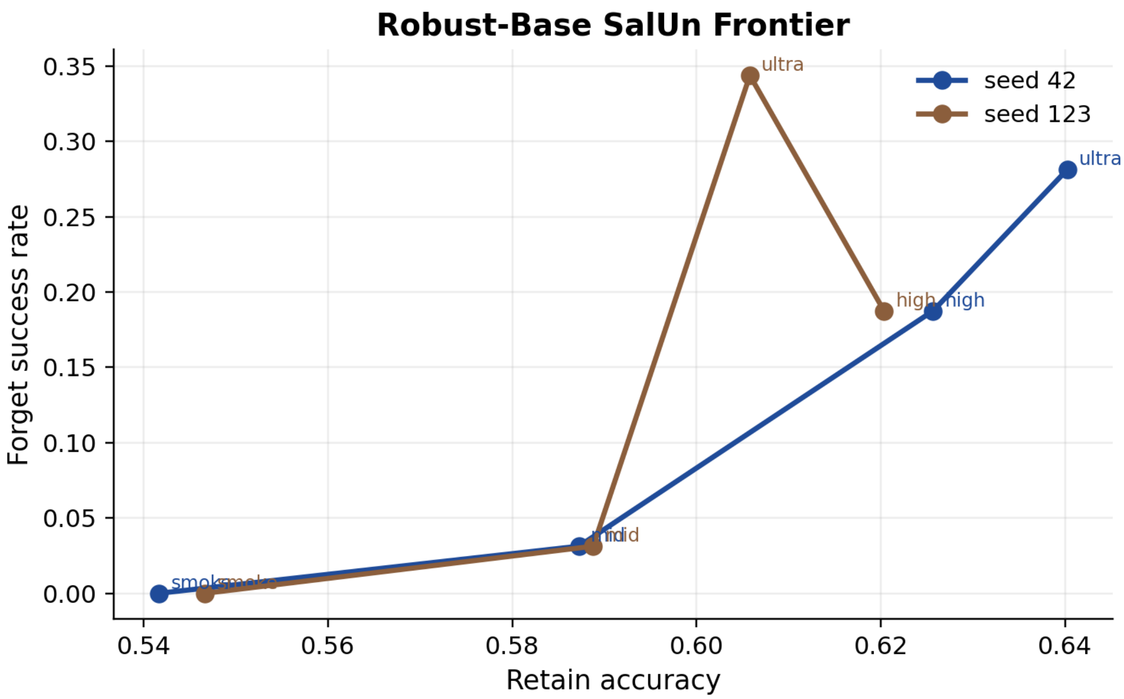

Figure 8: Robust-base SalUn operating-point frontier. Conditional recovery re-emerges as retain accuracy rises, confirming that robust-base training shifts the frontier rather than eliminating leakage.

F Appendix: Reproducibility Artifacts

The paper draws from repository summaries and generated figures in:

- `results/analysis/feature_subspace_tradeoff.md`
- `results/analysis/recovery_tradeoff_summary.md`
- `results/analysis/conditional_patch_summary.md`
- `results/analysis/paper_results_outline.md`
- `results/analysis/figures/`

All figures and tables in this paper are generated from JSON result files committed alongside the code, together with the audit scripts needed to reproduce each attack regime. Full checkpoint reproduction is heavier: the trained checkpoints live on a separate `checkpoints` branch (base, oracle, and several unlearn adapters across seeds 42, 123, and 456), supplemented by theory-era and stage-2 checkpoints that are regenerated locally from the training scripts. Not all eight benchmark methods are checkpoint-covered across all three seeds on the main branch; the tracked JSON artifacts, not the checkpoints, are the primary reproducibility surface.

G Appendix: Adaptation Protocol

A critical caveat for the fairness of this benchmark concerns CE-U, SGA, and BalDRO. These methods were originally proposed in the context of Large Language Models (LLMs). The implementations

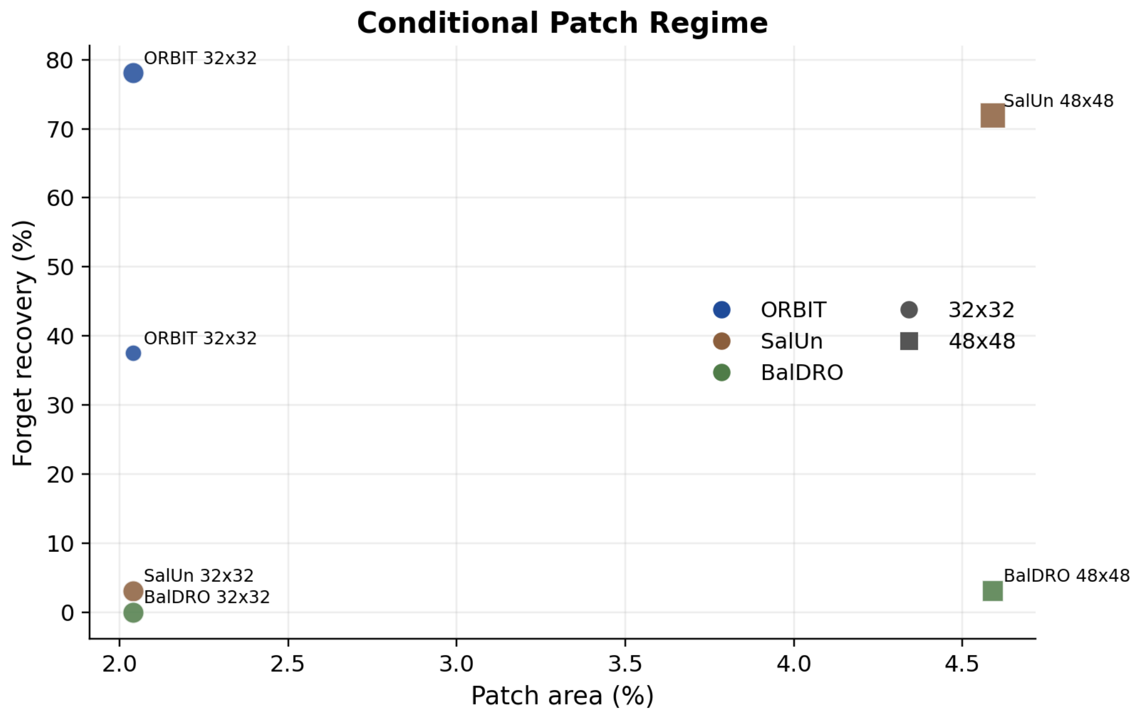

Figure 9: Conditional patch recovery rates across methods. ORBIT is the strongest patch-positive case; BalDRO and robust-base SalUn are effectively negative at 32×32.

evaluated in this paper are *objective-level adaptations* to the vision domain (CIFAR-10 classification with ViT architectures), not faithful reproductions of their original LLM settings or training dynamics. They are included to provide a diverse set of algorithmic loss structures for the robustness microscope to probe, not to formally evaluate the original authors' claims in the domain they were intended for.

H Appendix: Low-Shot Controls

The lower-shot prompt controls help separate real recovery from trivial prompt memorization. Figure 10 summarizes two useful trends: oracle controls remain flat near zero across the tracked shot counts, while KL and SCRUB can become vulnerable only in specific low-shot or high-shot regimes.

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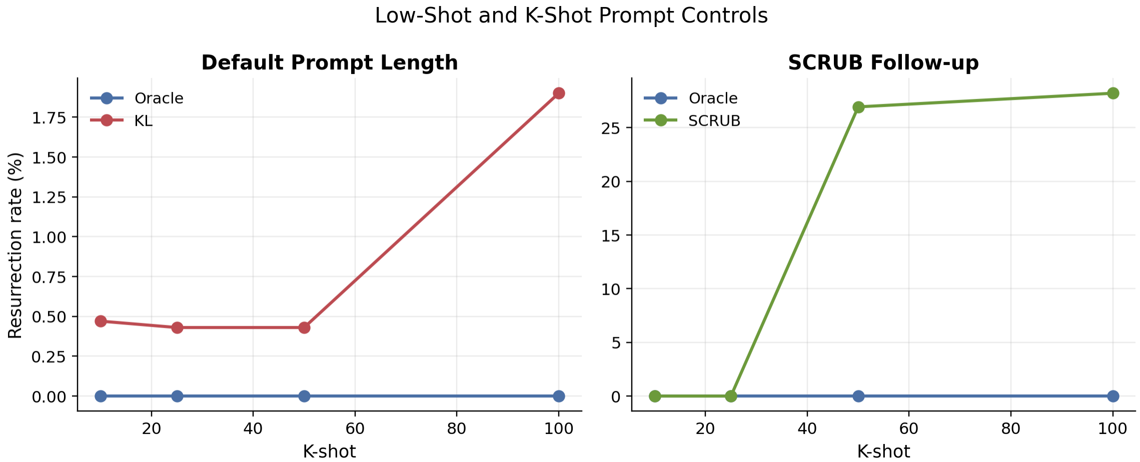

Figure 10: K-shot prompt controls. Oracle stays flat, KL remains weakly vulnerable, and SCRUB only becomes unstable in the larger-shot follow-up.

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